

Registration of Extent

Toward a Finite Symbolic Account of Measurement, Decoherence, and Dynamical
Stability

Foundational Measurement Series

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Abstract

This monograph develops a finite symbolic account of measurement in which registration, rather than a pre-assumed travelling entity, is taken as foundational. The argument begins from a simple distinction: the measured world is encountered as systems with extent, whereas the symbols used to record measurement—a click, a count, a coordinate, a binary value—do not preserve the extent of the physical interaction that produced them. Finite Symbolic Mechanics (FSM) therefore begins with extent, interaction, finite distinction, relational trajectory, stabilisation, and registration, and only afterwards admits symbolic representation.

From this starting point, the monograph reconstructs several familiar concepts of physics in a different order. Classical zero-extent points are replaced by finite distinguishable registrations at an Alphonic limit; isolated “beads” are then connected into relational strings that carry provenance and delay. Nonlinear phase-space reconstruction provides a means of describing stable dynamical organisation without requiring an infinitely divisible background space to be granted physical existence. Attractors are treated as stable organisations of continuing dynamics rather than as stable values, and averaging is shown to be a potentially destructive compression of trajectory.

The same method is then applied to the language of quantum mechanics. The path integral is unfolded as a symbolic compression of distinguishable trajectory contributions accumulated before final registration. Superposition is reconsidered as unresolved dynamical accessibility rather than the literal co-existence of symbolic outcomes. The two-slit experiment is reformulated as a single extended interacting system whose boundary conditions are altered by the aperture configuration. Decoherence is developed as a change of dynamical regime that occurs when a relational organisation stable at one scale is incorporated into a wider coupled system and transformed toward a coarser, robust registration.

The result is not presented as a completed replacement for quantum theory. It is a foundational measurement programme. Its purpose is to provide an admissible route from the measured world of extent to the symbolic world of recorded outcomes, and thereby to create a basis from which experimentally discriminating predictions may later be derived.

Contents

Prefatory Note	3
1 Registration Before Representation	4
1.1 The physical event and the symbolic result	5
2 The Registration Boundary	6
3 From Invisible Points to Finite Beads	8
4 Trajectory Before State	10
4.1 The string can fold	11
5 Attractors as Stable Organisation	12
6 Averaging as Trajectory Compression	14
6.1 Measurement as a many-to-one operation	15
7 Unfolding the Path Integral	16
8 The Problem of Quantum Language	18
8.1 Discrete registration and continuous reconstruction	18
9 Superposition as Unresolved Dynamical Accessibility	20
10 The Two-Slit Experiment as an Extended System	22
10.1 No assumed journey from A to B	23
11 Pattern of Registrations and Registration of a Pattern	24
12 The Click as an End Bead	26

13 System Enlargement and Decoherence	27
13.1 Nested enlargement	28
14 Measurement as Physical Coarse-Graining	29
14.1 Scale and loss of recoverable distinction	30
15 Collapse as Representational Discontinuity	31
16 One World of Extent	32
17 Registration as Foundational	33
17.1 The emerging trajectory	33
18 Two Methodological Routes	35
19 Predictions Must Follow the Model	36
Conclusion	37
A Working Vocabulary	38
B Provisional Statements for Later Testing	39

Prefatory Note

This work belongs to the foundational measurement series within Finite Symbolic Mechanics. Its direction is deliberately methodological. The aim is not to begin with the established ontology of classical or quantum physics and then reinterpret it. The aim is to begin earlier, at the point where an extended interaction becomes a registration and where that registration becomes a symbol.

This monograph develops the physical registration trajectory opened in *A Theory of Finite Representation, Part I*. That work establishes the distinction between physical, symbolic and bridge operations. Here the inquiry stays with the extended interaction: how it stabilises as a registration, what its symbolic endpoint preserves, and what may be lost when the trajectory is contracted into a count, coordinate or click. The proposals concerning quantum measurement arise from this registration programme; they are further constructions under its licence, rather than consequences already contained in the opening axiom.

The distinction is important because the symbolic result of measurement is often much smaller, cleaner, and more sharply bounded than the physical process that produced it. A detector click is represented as a single event. A CCD pixel is represented as a coordinate and a value. A photographic grain becomes a location in an image. A threshold crossing becomes a binary state. Yet the physical production of each of these registrations has spatial extent, temporal duration, material provenance, interaction history, and a surrounding dynamical context.

FSM therefore asks a prior question:

What is lost when an extended interactional event is compressed into a point-like symbol?

The monograph follows that question through nonlinear dynamics, phase-space reconstruction, path language, quantum measurement, the two-slit experiment, decoherence, and the emergence of robust macroscopic registration. The discussion is exploratory, but it is intended to terminate in a place from which experiment can begin.

Chapter 1

Registration Before Representation

The usual order of physical explanation is often representationally top-heavy. A mathematical state is defined, an equation evolves that state, and measurement is then introduced as the means by which the state becomes observable. In classical mechanics one may begin with a point in phase space. In quantum mechanics one may begin with a state vector in a Hilbert space. In each case the representational object is granted first and the measurement problem enters later.

FSM reverses this order.

The foundational sequence proposed here is

extent interaction finite distinction trajectory stabilisation registration symbol
--

The vertical bar is used relationally. It does not assert equality. It marks an admissible connection whose meaning depends upon provenance and the conditions of measurement.

The first commitment is to *extent*. A measured interaction occurs somewhere, lasts for some finite duration, and involves systems that are not dimensionless. The second commitment is to *interaction*. A registration does not arise from nothing; it is the result of one extended system changing in relation to another. The third commitment is to *finite distinction*. Whatever is recorded must be distinguishable at some finite limit. The fourth commitment is to *trajectory*: registrations occur in ordered relations, not as timeless isolated objects. The fifth is to *stabilisation*: the result must persist sufficiently for a subsequent system to register it. Only then does the symbolic representation arise.

This gives a different methodological principle:

Registration precedes representation.

The principle does not deny the usefulness of mathematical representation. It establishes provenance. A symbol is not treated as the event that produced it, and a mathematical space

containing symbols is not automatically treated as the physical space in which the underlying interaction occurred.

1.1 The physical event and the symbolic result

Consider a simple binary detector. A physical event changes the detector sufficiently for an electronic threshold to be crossed. The output is then represented as

$$0 \quad \text{or} \quad 1.$$

The registered marks 0 and 1 are finite instantiations in their own right. Under the detector's coding rule, each preserves a selected distinction in an extended interactional trajectory. Neither mark, by itself, reconstructs the extent, duration or sequence through which that distinction became stable.

The compression can be written schematically as

$$\text{extended interaction} \longrightarrow \text{stable physical registration} \longrightarrow \text{finite symbol}.$$

This is not a trivial bookkeeping distinction. Once the symbol has been produced, it can be manipulated algebraically as though its cleanliness and discreteness belonged to the original event. FSM insists that the provenance remain visible.

Chapter 2

The Registration Boundary

The boundary between physical registration and symbolic representation is not necessarily a boundary in nature. It is a boundary in a chain of interactions at which a particular description terminates and another begins.

A click in a detector can be treated as an endpoint because the experimenter chooses to regard it as the result. Physically, however, the click can become the beginning of another chain:

detector interaction $\rightarrow$ electronic amplification $\rightarrow$ speaker motion $\rightarrow$ air pressure change $\rightarrow$ ear $\rightarrow$ neural reg

A CCD behaves similarly:

CCD $\rightarrow$ ADC $\rightarrow$ memory $\rightarrow$ display $\rightarrow$ retina $\rightarrow$ neural system $\rightarrow \dots$

The “end” of the measurement is therefore relative to the chosen symbolic trajectory. What is terminal for one analysis is intermediate for another.

This suggests a generonic boundary: the point in a chosen measurement chain at which an analogue or materially distributed interaction is rendered as a finite symbol. The boundary is operationally real because instrumentation performs it, but it need not correspond to a fundamental ontological division in the world.

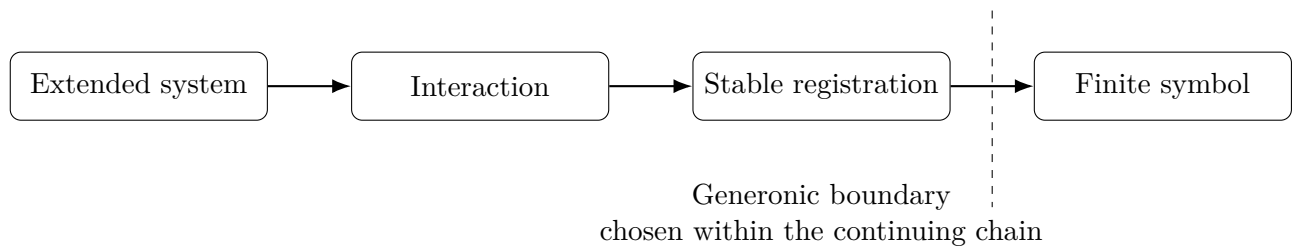


Figure 2.1: The registration chain. The symbolic boundary is chosen within a continuing physical interactional sequence.

Chapter 3

From Invisible Points to Finite Beads

Classical geometry begins comfortably with the point. The point has no extent, yet it is used to construct lines, surfaces, trajectories, coordinates, and continuous state spaces. Within the classical system this is mathematically coherent. Within FSM it is not an admissible physical primitive because a zero-extent point cannot itself be measured.

The replacement is not simply “discreteness” in the ordinary computational sense. FSM begins with a finite distinction at an Alphonic limit. We may depict such a registration as



The bead is a pedagogical image for a smallest admissible distinguishable registration in the current measurement system. It is not an invisible mathematical point. Its significance lies in the fact that it has provenance in a finite act of distinction.

Yet an isolated bead is not sufficient. A physical system is not reconstructed from disconnected registrations. The registrations must be related:

$$\bullet_0 \rightarrow \bullet_1 \rightarrow \bullet_2 \rightarrow \cdots \rightarrow \bullet_n.$$

This gives the bead string. The string is not assumed to be a continuous line hidden underneath the beads. It is a finite relation among distinguishable registrations. Its order matters. Its spacing may matter. Its delay may matter. Its history matters.

The transition from isolated beads to strings is foundational because it relocates complexity. Classical mathematics often increases descriptive power by permitting indefinitely finer subdivision. FSM instead holds the finite distinction and increases descriptive richness

relationally.

Complexity arises through relation rather than infinitesimal subdivision.

This change is small in notation but large in consequence. It permits dynamics without requiring a physically instantiated continuum of unmeasured intermediate points.

Chapter 4

Trajectory Before State

A measurement at one instant is an impoverished description of a dynamical system. A sequence of finite measurements contains more structure:

$$s_0, s_1, s_2, \dots, s_n.$$

The sequence can be unfolded into relational coordinates by delay embedding. For a scalar signal $s(t)$, a finite three-coordinate representation may be written

$$\mathbf{X}_t = [s(t), s(t - \tau), s(t - 2\tau)].$$

The coordinates are not granted as invisible axes populated by ideal points. They are finite symbolic relations among measured values. The embedding is therefore a reconstruction from registrations, not an ontological commitment to an independently existing mathematical space.

This is central to FSM's treatment of complex numbers. The usual notation

$$z = a + ib$$

is not taken to imply that a measurable system possesses a physically real coordinate and an ontologically imaginary coordinate. Complex notation can instead be interpreted as an efficient relational encoding, especially where phase and delay are involved. For suitable oscillatory relations,

$$x(t - \tau)$$

can carry the quadrature information ordinarily compressed into an i -weighted component.

The formal convenience of complex notation is retained, but the ontological inference is declined.

A finite trajectory is therefore prior to the isolated state. The object of interest is not merely

$$x_t,$$

but

$$x_{t-2\tau} \rightarrow x_{t-\tau} \rightarrow x_t.$$

The relation is the carrier of dynamical information.

4.1 The string can fold

Once delayed relations are represented in three coordinates, the symbolic string can fold through reconstructed phase space. Recurrence, divergence, convergence, cyclic behaviour, unstable regions, and attractor geometry can become visible in the finite representation.

This is important because it provides a route from finite registrations to geometric dynamical structure without first admitting a continuum as physically real.

Chapter 5

Attractors as Stable Organisation

An attractor is often described loosely as the state toward which a system settles. That phrasing is potentially misleading. A fixed point is one kind of attractor, but a limit cycle or strange attractor is not a fixed state. The system may continue changing indefinitely while remaining confined to a stable organisation of trajectories.

Let

$$\dot{\mathbf{x}} = F(\mathbf{x})$$

describe a nonlinear system. An attractor A may satisfy, schematically,

$$\mathbf{x}(t) \rightarrow A$$

for a set of initial conditions, while no individual $\mathbf{x}(t)$ becomes permanently fixed.

Thus the more useful formulation is:

An attractor is a stable organisation of continuing dynamics.

This distinction becomes decisive when measurement compresses a trajectory into a single value.

Consider the circular limit cycle

$$x(t) = \cos t, \quad y(t) = \sin t.$$

Its attractor is the unit circle,

$$x^2 + y^2 = 1.$$

The time averages are

$$\langle x \rangle = 0, \quad \langle y \rangle = 0.$$

The average point is therefore

$$(\langle x \rangle, \langle y \rangle) = (0, 0).$$

But the system never occupies that point.

The average is a symbolic construction derived from the stable dynamics. It is not itself the stable dynamics.

This example captures a general warning:

stable organisation $\neq$ stable value.
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Chapter 6

Averaging as Trajectory Compression

Averaging is often treated as a benign operation that removes unwanted fluctuation while preserving what matters. In nonlinear systems that assumption must be examined carefully.

A full trajectory may be represented as

$$\Gamma = \{\boldsymbol{x}(t)\}.$$

From that trajectory one may construct transition statistics, occupancy measures, moments, and finally a mean:

trajectory $\rightarrow$ transition statistics $\rightarrow$ occupancy distribution $\rightarrow$ moments $\rightarrow$ mean.

Each stage is a compression. Each removes relational detail.

For a nonlinear function F ,

$$F(\langle x \rangle) \neq \langle F(x) \rangle$$

in general. This inequality is not a technical inconvenience. It expresses the fact that nonlinear relations act on the trajectory before the average is taken. If the trajectory is compressed too early, the dynamics can be changed or destroyed.

This yields an important methodological statement for the present programme:

The trajectories must do their work before their symbolic compression.

The statement will recur when the path integral, interference, and registration are reconsidered.

6.1 Measurement as a many-to-one operation

An instrument may also implement a physical form of trajectory compression. Many distinct lower-scale interactions can produce the same detector state:

$$\gamma_1 \rightarrow R_0, \quad \gamma_2 \rightarrow R_0, \quad \gamma_3 \rightarrow R_0.$$

The registration R_0 is then represented by a symbol,

$$R_0 \rightarrow 0.$$

This gives two layers of compression:

dynamical compression $\rightarrow$ symbolic compression.

The first is performed by the physics of the instrument. The second is performed by the representational system.

Chapter 7

Unfolding the Path Integral

The phrase “path integral” is itself a compressed linguistic construction. FSM can unfold its words before accepting the ontology that often accompanies them.

A path is not a point. It implies an ordered relation:

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_n.$$

The noun “path” compresses a continuing sequence into a thing-like expression.

An integral likewise compresses an accumulation. A finite approximation exposes the operation:

$$\sum_{k=1}^N f(x_k) \Delta x_k.$$

One registration is taken, weighted, combined, followed by another, and another.

The phrase “path integral” can therefore be unfolded conceptually into

relational trajectory contributions accumulated before final compression.

The conventional path integral is more specific. Schematically,

$$K = \int \mathcal{D}[x] e^{iS[x]/\hbar}.$$

The integration is over paths rather than merely along one path. Within conventional quantum mechanics, the complex weighting carries relative phase, and the resulting amplitude is used to calculate probabilities.

FSM does not need to deny the operational success of this calculus. It asks what measurable relational structure the calculus may be compressing.

The terms can therefore be provisionally unfolded as follows:

path $\rightarrow$ finite distinguishable trajectory,
phase $\rightarrow$ finite relational delay or orientation,
complex value $\rightarrow$ compressed finite embedding,
integration $\rightarrow$ finite accumulation of relational contributions,
all paths $\rightarrow$ available distinguishable trajectory classes.

The last line is especially important. “All paths” invokes a continuous infinity in the conventional formal construction. Yet any actual symbolic calculation and any actual experiment has finite resolution. If two paths cannot be distinguished below the admissible resolution, then within FSM they cannot be treated as separately measured trajectories.

The question is therefore changed. Instead of asking whether an infinity of paths physically exists, FSM asks:

What finite relational dynamics are being compressed by the notation?

Chapter 8

The Problem of Quantum Language

Quantum experiments are marked by finite registrations: clicks, spots, threshold crossings, spectral lines, discrete detector states, and binary outputs. Yet the standard mathematical description is built using structures that permit indefinitely fine distinction:

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathcal{H}, \quad \psi, \quad \int \mathcal{D}[x].$$

The difficulty for FSM is not that these structures are mathematically ineffective. The difficulty is that the language used to describe them can transfer representational properties back into the physical world.

A state is said to “live in” Hilbert space. A system is said to “be in” superposition. A wavefunction is said to “collapse”. A photon is said to travel one path, both paths, or no definite path. Mathematical relations begin to sound like physical locations and behaviours.

FSM treats this as a provenance problem.

The measured event precedes the symbolic construction. The symbolic construction may be extremely effective, but its success does not itself prove that the world is made of the mathematical primitives used to represent the event.

The guiding question remains:

What measured trajectory preceded this symbol?

8.1 Discrete registration and continuous reconstruction

Quantum mechanics recognises discrete outcomes while representing them within continuous mathematical spaces. FSM regards that move as a historical and mathematical construction requiring justification rather than as a primitive fact.

The tension can be written schematically:

finite registration $\nRightarrow$ physically instantiated infinitely divisible state space.

A continuous state space may be useful. It is not automatically admissible as the substrate of the measured world.

Chapter 9

Superposition as Unresolved Dynamical Accessibility

A two-state quantum system is conventionally represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The notation is operationally powerful, but it can generate difficult language. If the output symbols 0 and 1 are projected backwards into the pre-registration world, one is tempted to say that the physical system is somehow both outcomes at once.

FSM declines that projection.

Suppose instead that a wider nonlinear system possesses two eventual registration basins,

$$B_0 \quad \text{and} \quad B_1,$$

separated by a finite dynamically sensitive region,

$$\Sigma$$

.

The finite region Σ

replaces the idea of an infinitely thin measured separatrix. A dynamical trajectory within this region may retain access to more than one future registration without itself being either symbolic outcome.

Thus

$$B_0 \mid \Sigma$$

$$\mid B_1$$

does not mean

$$0 + 1.$$

It means that the wider system has not yet stabilised into either terminal symbolic basin.

The corresponding registration chain is

$$\Sigma$$

$$\rightarrow \begin{cases} B_0 \rightarrow 0, \\ B_1 \rightarrow 1. \end{cases}$$

This gives a less ontologically loaded description of the same experimental circumstance:

Neither symbolic registration has yet stabilised.

The proposal is not that every quantum superposition has already been explained by a saddle point. The saddle is the simplest geometric model of unresolved dynamical accessibility. The wider research question is whether finite nonlinear dynamical organisations can be reconstructed from measured quantum-device trajectories in a way that makes conventional superposition notation a compressed representation rather than a primitive ontology.

Chapter 10

The Two-Slit Experiment as an Extended System

The two-slit experiment is often narrated as the journey of an entity from source to detector. A photon or particle is prepared at one region, travels toward two openings, and is eventually registered at another. Once the travelling entity is granted as the primitive, the explanatory problem becomes the path it took.

FSM begins elsewhere.

Let the experimental arrangement be an extended system

$$\mathcal{S} = \{\text{source, apertures, materials, surroundings, detector, } \dots\}.$$

The experiment with both apertures open is one configured system,

$$\mathcal{S}_{AB}.$$

Closing one aperture produces a different system,

$$\mathcal{S}_A,$$

and closing the other produces

$$\mathcal{S}_B.$$

Therefore,

$$\boxed{\mathcal{S}_{AB} \neq \mathcal{S}_A \neq \mathcal{S}_B.}$$

This statement is elementary but foundational. Closing a slit does not merely remove

one route from a pre-existing set of paths available to a traveller. It changes the boundary conditions and therefore the wider interacting system.

If the system is nonlinear, there is no general reason to expect the registration behaviour of the two-aperture configuration to be the sum of the registration behaviours of the two separate one-aperture systems:

$$\mathcal{R}(\mathcal{S}_{AB}) \neq \mathcal{R}(\mathcal{S}_A) + \mathcal{R}(\mathcal{S}_B)$$

in general.

The experimentally observed non-additivity can therefore be investigated as a property of the configured system before it is interpreted as evidence for a travelling object simultaneously taking multiple paths.

This is a methodological reversal. It does not yet reproduce the interference distribution. It changes the level at which the explanation is sought.

10.1 No assumed journey from A to B

Within FSM, an unmeasured line from source to detector is not admitted merely because classical geometry can draw it. The best available representation is a finite chain of registrations:

$$\bullet_0 \rightarrow \bullet_1 \rightarrow \cdots \rightarrow \bullet_n.$$

The final bead may be a detector click, a CCD event, a chemical change in photographic material, or a retinal interaction. The intervening relation is reconstructed from the best finite evidence available. It is not silently replaced by an invisible continuous trajectory.

Chapter 11

Pattern of Registrations and Registration of a Pattern

The two-slit experiment also exposes a linguistic compression that is easy to miss.

In a low-flux accumulation experiment, the detector does not first register a smooth sinusoidal intensity field. It registers local events:

$$r_1, r_2, \dots, r_N.$$

These events are subsequently binned or accumulated spatially:

$$N(x).$$

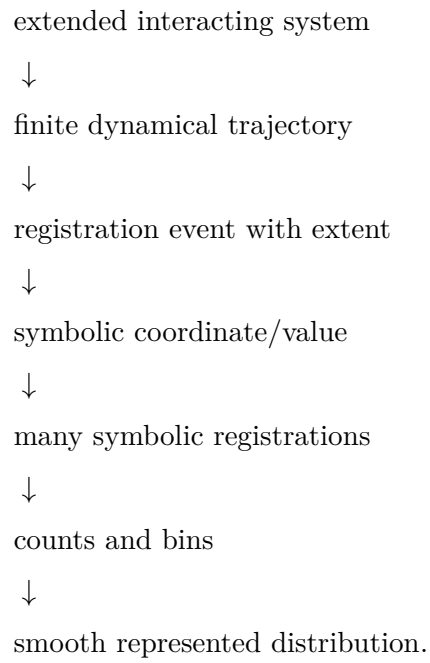
A smooth distribution may then be fitted, plotted, or visually inferred.

The distinction is:

pattern of registrations $\neq$ registration of a pattern.
--

This does not by itself determine the underlying dynamics. It does prevent a representational reversal in which the smooth curve constructed from many registrations is projected backwards and treated automatically as the thing that travelled through the apparatus.

The chain is instead kept explicit:



The smooth pattern is real as a representation of the accumulated data. It need not be admitted as a primitive travelling object.

Chapter 12

The Click as an End Bead

The detector click is a useful boundary object because it shows how a continuing physical interaction becomes a terminated symbolic trajectory.

Let a chosen measurement chain be represented as

$$\cdots \rightarrow \bullet_{n-2} \rightarrow \bullet_{n-1} \rightarrow \boxed{\bullet_{\text{click}}}.$$

The click is the end bead relative to that particular analysis. It is not an absolute physical endpoint.

The same event may become the first bead in another chain:

$$\boxed{\bullet_{\text{click}}} \rightarrow \bullet_{\text{electronics}} \rightarrow \bullet_{\text{auditory}} \rightarrow \bullet_{\text{neural}} \rightarrow \cdots.$$

This gives a useful interpretation of measurement: measurement is not necessarily where the world stops changing. It is where a selected trajectory is considered sufficiently stable to be registered and represented.

The phrase “this is the best we can do” is important here. FSM does not claim access to a hidden continuous path beneath the bead chain. The bead chain is the admissible finite reconstruction produced from measurement.

Chapter 13

System Enlargement and Decoherence

The usual quantum language places decoherence near the boundary between a delicate quantum system and a larger measuring environment. FSM retains the empirical fact that coupling to a wider system changes what can be stably registered, but reconstructs the process in nonlinear dynamical terms.

Begin with a system

$$S_A$$

having a stable or metastable organisation

$$A(S_A).$$

Now couple it to another system:

$$S_A|S_B \longrightarrow S_{AB}.$$

The enlarged system is not merely the old system plus a passive observer. It has different coupling terms, different constraints, different accessible trajectories, and therefore potentially different attractors.

There is no general requirement that

$$A(S_A)$$

remain an attractor of

$$S_{AB}.$$

Indeed,

$$A(S_A) \not\Rightarrow A(S_{AB}).$$

The act of combining systems may therefore make a previously stable relational organisation unstable.

This gives the provisional FSM decoherence conjecture:

A dynamically stable relational organisation at one interactional scale may become unstable when incorporated into a wider coupled system, whose dynamics then converge toward a coarser stable organisation capable of producing a finite registration.

Decoherence, in this account, is not a mysterious disappearance of a physically real superposition in an invisible space. It is a change of dynamical regime associated with system enlargement across scale.

13.1 Nested enlargement

The enlargement can continue:

$$S_A \rightarrow S_{AB} \rightarrow S_{ABC} \rightarrow S_{ABCD} \rightarrow \cdots$$

Each stage may possess a different phase-space organisation.

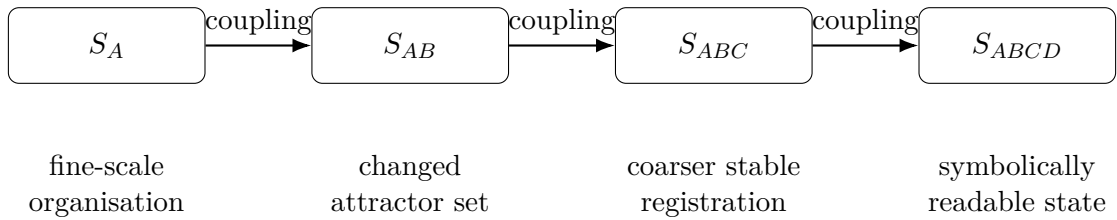


Figure 13.1: System enlargement changes the dynamical system under consideration. Decoherence is investigated as a regime change within this nested sequence.

The conventional question “where does the quantum system become classical?” may therefore be badly posed. The system is repeatedly changed by coupling. There may be no single ontological boundary, only successive dynamical transformations across scale.

Chapter 14

Measurement as Physical Coarse-Graining

The detector does not merely observe lower-scale detail. It is designed to suppress much of that detail and produce robust macroscopic states.

Suppose several fine trajectories enter the detector:

$$\gamma_1, \gamma_2, \gamma_3, \gamma_4.$$

The detector may map them into only two stable macroscopic registrations:

$$\gamma_1, \gamma_2, \gamma_3 \rightarrow R_0, \quad \gamma_4 \rightarrow R_1.$$

This is a physical many-to-one map. The instrument has performed a coarse-graining through its own dynamics.

The registrations may then be symbolised:

$$R_0 \rightarrow 0, \quad R_1 \rightarrow 1.$$

The full sequence is therefore

$$\{\gamma_i\} \rightarrow R_j \rightarrow j.$$

The first arrow is a dynamical compression across scale. The second arrow is a symbolic compression.

This distinction matters because it relocates part of the measurement problem into the actual design and dynamics of instrumentation. A detector is not simply a transparent window onto a pre-existing point-like state. It is a nonlinear dynamical system whose stable

macroscopic outputs are deliberately engineered.

14.1 Scale and loss of recoverable distinction

Let ℓ denote a fine distinguishable scale and L the characteristic scale of a macroscopic registration, with

$$L \gg \ell$$

A large number of lower-scale relational configurations may become indistinguishable once mapped into the same L -scale state.

Thus the loss of fine structure need not be interpreted as the metaphysical destruction of information. It is the loss of recoverable distinction within the enlarged system and chosen representation.

This gives decoherence a natural connection with registration: the system becomes stable enough at the larger scale precisely because many lower-scale distinctions no longer alter the macroscopic outcome.

Chapter 15

Collapse as Representational Discontinuity

The language of collapse presents a dramatic transition:

$$|\psi\rangle \rightarrow |0\rangle.$$

In the standard formalism, the transition marks the change from a pre-measurement state description to a definite registered outcome. Within FSM, the same experimental sequence can be represented without assuming that an instantaneous physical jump corresponds directly to the symbolic discontinuity.

The interacting sequence may instead be

$$S_A \rightarrow S_{AB} \rightarrow S_{ABC} \rightarrow \cdots \rightarrow R_0 \rightarrow 0.$$

Each arrow may contain finite dynamical change. The apparent discontinuity arises because a high-dimensional, extended interactional trajectory is compared with the compact symbol produced at the end.

Thus one possible FSM conjecture is:

Collapse is at least partly a representational discontinuity generated when an extended dynamical process is compressed into a finite symbolic state.

The phrase “at least partly” is essential. The conjecture does not assume in advance that all experimentally observed discontinuity is representational. It establishes a framework in which the physical transition and the symbolic transition can be investigated separately.

Chapter 16

One World of Extent

The analysis now permits a different formulation of the quantum/classical problem.

Rather than asking

Where does the quantum world become classical?

FSM asks

How does a fine-scale relational dynamical organisation transform as it is progressively coupled into wider systems until a stable registration becomes possible?

The latter question requires only one world of extent.

There are systems within systems, interacting across scales:

$$S_1 \rightarrow S_{12} \rightarrow S_{123} \rightarrow \cdots .$$

There are changes in stability, attractor structure, accessible trajectory, and registration.

There is no need at the foundational level to place an invisible mathematical wall between quantum and classical ontology. The distinction may instead emerge from the representational consequences of scale, coupling, and stabilisation.

This does not make the experimental phenomena disappear. It relocates them into a framework that begins with measurable extent rather than with an abstract state space.

Chapter 17

Registration as Foundational

The argument can now be condensed into a provisional FSM registration principle:

Physical measurement begins with interacting systems of extent.
Interaction produces finite distinguishable changes.
Relations among these distinctions form trajectories.
Coupling changes the dynamical organisation of the wider system.
Stable macroscopic organisation produces registration.
Registration is subsequently compressed into a symbol.

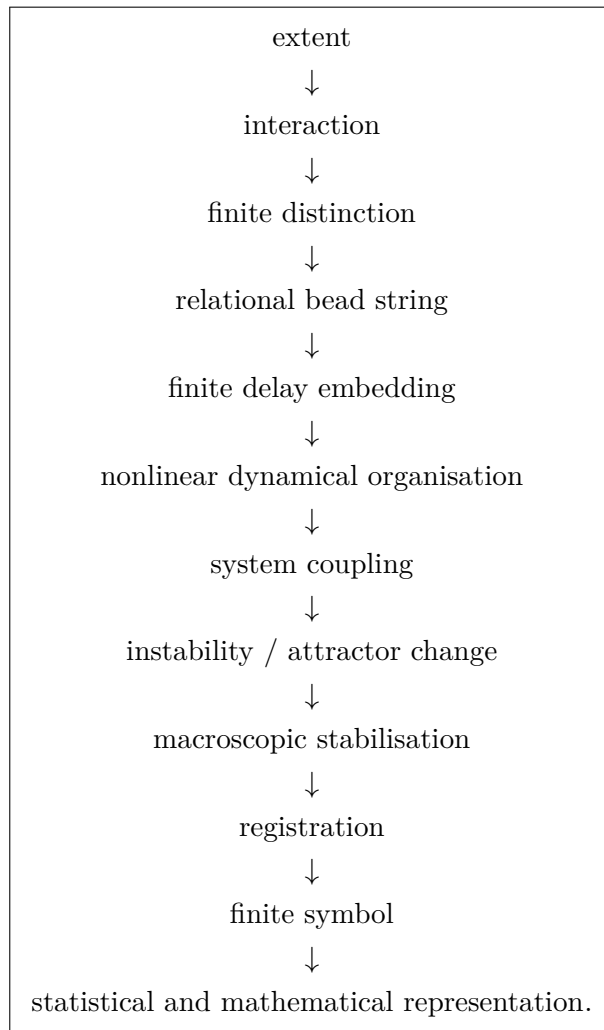
The order matters.

If representation is placed first, then physical explanation is tempted to inherit the ontology of the representation. If registration is placed first, then every symbolic construction is required to retain provenance back to a finite interactional event.

This is the sense in which registration is foundational.

17.1 The emerging trajectory

The full sequence developed in this monograph is



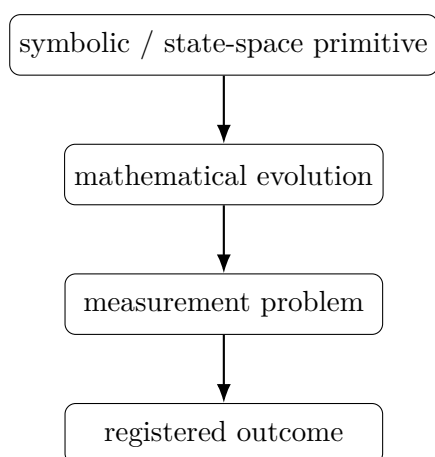
The route is intentionally opposite to one that begins with a mathematical state and then seeks to explain how it becomes a measurement.

Chapter 18

Two Methodological Routes

The difference can be displayed directly.

Conventional representational route



FSM measurement route

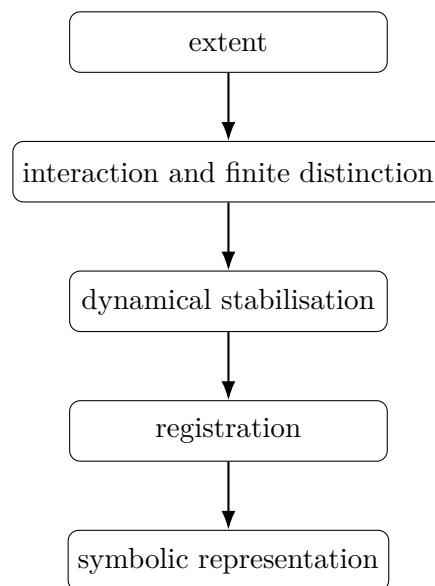


Figure 18.1: Two different starting points. FSM delays symbolic ontology until after finite registration.

The comparison should not be read as a caricature of all existing physics. Modern experimental practice already depends upon finite detectors, thresholding, digitisation, calibration, and dynamical apparatus. The point is that the foundational mathematical language often places those operations downstream of an already specified state space.

FSM brings them upstream.

Chapter 19

Predictions Must Follow the Model

The present work stops deliberately before specifying the experimental programme.

A new language is not sufficient merely because it appears clearer. If registration is genuinely foundational, and if a two-slit or related experiment is more accurately modelled as an extended nonlinear system rather than as the transit of a primitive wave or photon, then the framework should eventually produce a distinguishable expectation under some controllable experimental condition.

The methodological sequence should be

$\text{model} \rightarrow \text{prediction} \rightarrow \text{experimental design} \rightarrow \text{registration} \rightarrow \text{comparison}.$

This is where the next stage of the foundational measurement series begins.

The experimental task is not to prove a philosophy. It is to identify a condition under which the registration-based nonlinear model and a conventional travelling-entity model do not make the same expectation.

Possible experimental variables will need to be chosen from quantities that alter the wider interacting system rather than merely relabel its symbolic outputs. These may include geometry, separation, material boundary conditions, detector coupling, scale, delay, interaction density, or controlled changes in the registration chain. The actual prediction should precede the detailed instrument design.

The present monograph therefore terminates with a methodological commitment:

The purpose of this work is not to offer another interpretation of an already accepted quantum ontology, but to construct a finite admissible starting point from which an experimentally distinguishable prediction may be made.

Conclusion

The central issue examined here is the registration of extent.

The measured world is encountered through systems that possess extent and through interactions that unfold over finite intervals. Instruments transform those interactions into stable registrations. Those registrations are then compressed into symbols that no longer preserve the extent of the physical events from which they arose.

Classical and quantum mathematical languages can obscure this transition because they begin comfortably with symbols that behave as ideal points, coordinates, states, or vectors in mathematical spaces. Once the symbolic framework has been granted, language can easily project its primitives backwards into the measured world.

FSM takes the opposite route.

It begins with extent. It admits finite distinction. It connects distinguishable registrations into relational strings. It uses delay and phase-space reconstruction to recover dynamical organisation. It treats attractors as stable organisations rather than stable values. It treats averaging as a potentially destructive compression of trajectory. It unfolds the path integral as a calculus that retains trajectory distinctions until a later stage of composition. It reconsiders superposition as unresolved dynamical accessibility. It treats the two-slit apparatus as a single configured system whose boundary conditions change when an aperture is altered. It interprets detector registration as physical coarse-graining across scale. And it proposes decoherence as a change of dynamical regime produced when a stable relational organisation is incorporated into a wider system with different attractors.

The resulting sequence is economical:

extent → interaction → distinction → trajectory → stabilisation → registration → symbol.
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This sequence is the foundation on which the next work can proceed.

The next question is experimental.

If registration rather than travelling wave or photon is taken as foundational, what measurable prediction follows?

Appendix A

Working Vocabulary

Alphonic limit. The smallest admissible finite distinction in the chosen measurement and representational system. It marks a limit of distinguishability, not an invitation to continue symbolic subdivision without measurement.

Bead. An informal visual term for a finite distinguishable registration at the Alphonic limit or other explicitly stated resolution.

Bead string. An ordered relational sequence of finite registrations. The string does not imply an unmeasured continuous line beneath the registrations.

Registration. A physically stable change in one system produced through interaction with another system, sufficiently persistent to become available for subsequent measurement or symbolic encoding.

Symbol. A finite representational mark assigned to a registration. The symbol need not preserve the physical extent, duration, or internal dynamics of the registration event.

Generonic boundary. The chosen operational boundary at which a continuing physical interactional chain is rendered into a finite symbolic form.

Dynamical compression. A many-to-one physical mapping in which multiple lower-scale trajectories converge into a smaller set of robust macroscopic registrations.

Symbolic compression. The subsequent mapping of a stable registration into a finite symbolic state such as a count, coordinate, label, or binary value.

Unresolved dynamical accessibility. A condition in which the wider system has not yet stabilised into one of several available registration basins. This provides an FSM route for investigating phenomena ordinarily described using superposition language.

FSM decoherence conjecture. The proposal that a stable relational organisation at one interactional scale can become unstable when incorporated into a wider coupled system, whose dynamics then converge toward a coarser stable organisation capable of producing a finite registration.

Appendix B

Provisional Statements for Later Testing

The following statements are included as research markers rather than as established results.

1. A detector registration is better modelled as the stable outcome of an extended nonlinear interaction than as the direct revelation of an ideal point-state.
2. A change in experimental boundary conditions alters the whole configured system and may alter its attractor structure.
3. A two-slit registration pattern should therefore be investigated first as a property of the whole extended system, not as proof of a primitive travelling entity taking more than one path.
4. Apparent collapse may contain a representational discontinuity introduced by the mapping from extended dynamics to a compact symbol.
5. Decoherence may be experimentally reconstructible as a transition between dynamical organisations across system enlargement and scale.
6. If the registration model is foundational rather than merely interpretive, at least one experimental condition should eventually be found under which it predicts a registration distribution or dynamical transition distinguishable from a conventional travelling-entity account.